Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/22/2025
PI Name:	Jinkerson, Robert Ph.D.		
Project Title:	Evaluation of Small Plants for Agriculture in Confined Environments (SPACE) Tomatoes for Space Flight Applications		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	92507	**Congressional District:**	41
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	03/01/2021	**End Date:**	02/28/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 02/28/2026 per E. Blancaflor/NASA-KSC (Ed., 7/22/25). NOTE: End date changed to 02/28/2025 per G. Massa/NASA-KSC (Ed., 1/23/25).		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Orozco-Cardenas, Martha Ph.D. (University of California, Riverside)		
Grant/Contract No.:	80NSSC21K0578		
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Performance Goal Text:			

Task Description:	Plants can have a tremendously beneficial impact on long spaceflights if some biological hurdles can be overcome. Plants on extended space expeditions can provide a fresh source of food and nutrients, CO2 uptake capacity, and behavioral health benefits to crewmembers. Most fruit and vegetable bearing plants, however, are too large and produce too much inedible biomass to be part of a bioregenerative life support system. We have recently increased the developmental rate and harvest index of tomato plants dramatically through genetic engineering. These plants rapidly progress through their developmental cycle to produce fruit, minimizing their size and producing little non-fruit biomass. We are calling this extremely dwarf phenotype Small Plants for Agriculture in Confined Environments (SPACE) tomatoes. These plants have been evaluated on Earth but it is not known how this dwarf plant phenotype will manifest in a microgravity environment. This project aims to (1) cultivate SPACE tomato plants in the Advanced Plant Habitat aboard the International Space Station (ISS); (2) determine the physiological response of these plants to grow in microgravity; (3) evaluate fruit grown on the ISS for yield, nutrient levels, and microbial loading; and (4) complete a full life cycle of the SPACE tomatoes aboard the ISS and determine if growing tomatoes seed-to-seed in space alters fruit yields of progeny. The SPACE phenotype is innovative and potentially transformative for spaceflight and ground-based controlled environment agriculture. The SPACE tomato is uniquely suited for environments where physical space is limited. Relatively little research has been done on producing plant traits such as SPACE tomatoes because they serve little agricultural importance in present-day ground-based agricultural systems. Most other mutations that produce plant dwarfs largely keep the proportion of leafy, un-edible material to edible fruit the same. NASA has investigated several dwarf tomatoes but none have been as extreme as the SPACE tomato. The SPACE trait forces the plant quickly through its developmental cycles to produce fruit without the necessity to develop the whole plant. This results in profoundly small plants that produce fruit that is a high fraction of their biomass. This work will determine how well suited SPACE tomatoes are for a bioregenerative life support system. Morphological data of SPACE tomato plants grown in microgravity will also inform future plant genetic engineering strategies on how to create other dwarf plant varieties that are ideally suited for growth on a spacecraft in microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The SPACE phenotype is innovative and potentially transformative for spaceflight and ground-based controlled environment agriculture. The SPACE tomato is uniquely suited for environments where physical space is limited. Relatively little research has been done on producing plant traits such as SPACE tomatoes because they serve little agricultural importance in present-day ground-based agricultural systems. Most other mutations that produce dwarf plants largely keep the proportion of leafy, un-edible material to edible fruit the same. NASA has investigated several dwarf tomatoes but none have been as extreme as the SPACE tomato. The SPACE trait forces the plant quickly through its developmental cycles to produce fruit without the necessity to develop the whole plant. This results in profoundly small plants that produce fruit that is a high fraction of their biomass. This work will determine how well suited SPACE tomatoes are for a bioregenerative life support system. Morphological data of SPACE tomato plants grown in microgravity will also inform future plant genetic engineering strategies on how to create other dwarf plant varieties that are ideally suited for growth in physically confined locations, such as on Earth, in controlled environment agriculture. Data collected on SPACE tomato plant growth, yield, and performance during this project can be used to estimate similar metrics for on-Earth production of SPACE tomatoes in controlled environment agriculture setups. Additionally, optimized culture conditions for SPACE tomato plant growth, such as light, temperature, humidity, and fertilizer levels, can all be used as starting points for tomato growth in on-Earth controlled environment agriculture, such as in vertical farms.
Task Progress:	Over the past year, the project achieved several major milestones that substantially advance readiness for spaceflight. Most notably, the Science Verification Test (SVT) was successfully completed at NASA Kennedy Space Center using the Advanced Plant Habitat (APH) ground unit, validating the current Experiment Requirements Document (ERD) and demonstrating robust growth, nominal operations, and successful seed-to-seed propagation for both wild-type (WT) and SPACE tomato genotypes. In parallel, extensive data were generated to support FDA review of SPACE tomato fruit for astronaut consumption, including α-tomatine quantification and total lysine content determination, with results indicating no safety concerns relative to wild-type or commercially available tomatoes. Additionally, an independent spaceflight experiment (NIGHT mission) demonstrated high germination rates and early growth of SPACE tomatoes in microgravity aboard the International Space Station (ISS), providing complementary validation of seed quality and spaceflight viability. Collectively, these accomplishments position the project strongly for progression to an Experiment Verification Test (EVT) or a direct flight experiment and represent significant progress toward deploying SPACE tomatoes as a consumable crop for future space missions.
Bibliography Type:	Description: (Last Updated: 07/05/2023)